When Does Conformer Geometry Help? Complementarity of 3D Ensemble Statistics and 2D Fingerprints for Molecular Property Prediction

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Abstract

We study when three-dimensional conformer ensemble statistics complement two-dimensional molecular fingerprints for property prediction. Our central finding is one of *selective complementarity*: while Morgan fingerprints with XGBoost outperform all neural conformer methods on every benchmark, hybrid features combining fingerprints with conformer statistics yield 9.9% RMSE reduction on ESOL, 3.9% on FreeSolv, and 4.2% on QM9-HOMO—establishing conformer statistics as complementary descriptors for solvation-dependent properties. Using Distribution Kernel Operators (DKOs), we extract mean and covariance features from conformer ensembles and evaluate 13 model configurations across 14 regression targets. Hybrid features combining fingerprints with enhanced physicochemical 3D descriptors (PMI, SASA, USR) achieve RMSE 1.000 on ESOL, a 34% improvement over fingerprints alone. We establish a performance hierarchy for solvation prediction: end-to-end 3D GNNs (SchNet) and hybrid FP+3D features surpass fingerprint baselines by 21–34%, while fingerprints outperform all neural conformer ensemble methods by 8–44%. This complementarity is selective: an empirical property taxonomy (solvation vs. steric vs. electronic) suggests when conformer features add value. Mechanistic analysis via mutual information and feature attribution reveals that conformer mean features carry 2–8× more information per feature than fingerprint bits, while covariance features contribute <2% of total model signal. Among neural architectures, learned gating for mean/covariance fusion significantly outperforms attention ($p < 0.001$), and five simple covariance summary statistics outperform complex representations.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Modeling and simulation*; • **Applied computing** → Life and medical sciences.

Keywords

molecular property prediction, conformer ensembles, distribution kernel operators, fingerprints, 3D descriptors, covariance representations

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1 Introduction

Molecular property prediction is a cornerstone of computational chemistry and drug discovery. While graph neural networks (GNNs) operating on 2D molecular graphs have achieved remarkable success [11, 38], a molecule's biological activity fundamentally depends on its three-dimensional geometry. Small molecules adopt multiple low-energy conformations in solution, and properties such as solvation free energy and lipophilicity are Boltzmann-weighted averages over this conformational ensemble [12].

Accurate prediction of solvation properties is particularly critical for ADMET (absorption, distribution, metabolism, excretion, and toxicity) profiling in early-stage drug discovery [14, 21], where experimental measurement is expensive and computational screening must efficiently prioritize candidates (Figure 1).

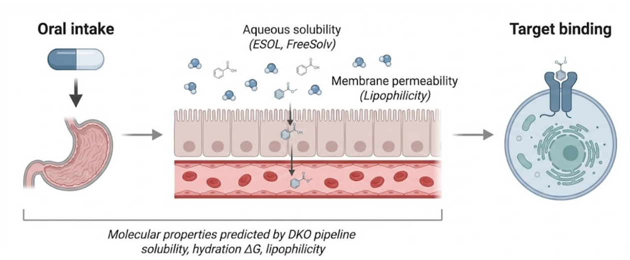

Figure 1: Mechanistic context for molecular property prediction. Oral drugs must dissolve (aqueous solubility), cross epithelial membranes (lipophilicity), and reach their cellular target. The DKO pipeline predicts these solvation-dependent properties from conformer ensemble statistics. Created with BioRender.com [4].

This has motivated a growing body of work on conformer ensemble methods [1, 7, 34] and 3D-aware GNNs [9, 32, 33], which incorporate 3D geometry for property prediction. However, a fundamental question remains: *when does explicit 3D conformer geometry provide signal beyond what 2D molecular descriptors already capture?*

We address this question through a systematic benchmark of 13 model configurations (7 DKO variants and 6 baselines) across 14 regression targets. Our approach extracts first-order (mean $\boldsymbol{\mu}$) and second-order (covariance Σ) statistics from conformer features and evaluates multiple fusion architectures. **Scope:** this study focuses on regression tasks; preliminary experiments on classification (BACE, BBBP) showed degenerate DKO behavior, suggesting architectural adaptation is needed for discrete outputs. Our key contributions are:

(1) **DKO framework**: A Distribution Kernel Operator architecture with explicit $\boldsymbol{\mu}/\Sigma$ modeling, 7 covariance representation variants, and 28 enhanced 3D shape descriptors, including a gated fusion mechanism that significantly outperforms attention ($p < 0.001$).
(2) **Comprehensive benchmark**: 13 model configurations evaluated across 14 regression targets spanning MoleculeNet, MARCEL, and QM9, plus SchNet 3D GNN baseline, 28 enhanced physicochemical descriptors, and conformer count ablation, with 10-seed statistical validation.
(3) **Selective complementarity finding**: While fingerprints alone beat all neural conformer methods, hybrid features improve predictions on solvation-related tasks by up to 9.9%. End-to-end 3D learning (SchNet) surpasses all pre-computed feature methods by 21–33%.
(4) **Mechanistic understanding**: Quantitative feature attribution and mutual information analysis reveal *why* fingerprints dominate and *why* hybrids help selectively. An empirical property taxonomy (solvation vs. steric vs. electronic) suggests when conformer features add value (Figure 4).

2 Related Work

Conformer ensemble methods. The MARCEL benchmark [1] established standardized evaluation for conformer ensemble learning on Kraken steric descriptors, bond dissociation energies, and electronic properties. GeoMol [7] generates conformers via end-to-end learning, while 3D Infomax [34] uses contrastive 3D pre-training to enrich 2D representations. Alternative conformer generation methods include OMEGA [13] and CREST [26]; we use ETKDG for consistency with the MARCEL protocol. These approaches typically aggregate conformer features via mean pooling or attention, discarding distributional information.

3D molecular representations. SchNet [32] introduced continuous-filter convolutions for modeling quantum interactions from 3D coordinates. DimeNet [9] and its successor DimeNet++ [8] incorporate directional message passing using bond angles. EGNN [31] introduces E(n)-equivariant message passing, while PaiNN [33] achieves equivariant message passing for tensorial properties. SphereNet [22] extends to spherical coordinates, GemNet [16] uses geometric message passing with universal approximation guarantees, and Equiformer [20] combines equivariant attention with transformer architectures. Uni-Mol [40] provides large-scale 3D pre-training, and TorchMD-NET [35] combines equivariant transformers with molecular dynamics potentials. These methods operate on single conformers or equilibrium geometries; our work instead models the *ensemble distribution* via summary statistics.

Molecular fingerprints and classical ML. Extended-connectivity fingerprints [29] remain the dominant representation in cheminformatics. Combined with gradient-boosted trees [5], they provide baselines that neural methods frequently fail to surpass [37]. Our work quantifies this gap and identifies where conformer features add complementary value.

Set aggregation architectures. Processing unordered conformer sets requires permutation-invariant architectures. DeepSets [39] provides the theoretical foundation, while Set Transformer [19] introduces attention-based aggregation. We compare these against our DKO framework, which models the distribution's first and second moments explicitly.

Kernel methods in molecular ML. Kernel methods have a long history in molecular property prediction [30]. Our DKO builds on distributional kernels that compare molecules via their conformer distributions, using a learned positive semi-definite kernel $\mathbf{K} = \mathbf{L}\mathbf{L}^\top$.

3 Methods

Figure 2 provides an overview of the DKO pipeline and hybrid approach.

3.1 Conformer Generation and Feature Extraction

For each molecule, we generate $n = 50$ conformers using RDKit's [18] ETKDG (Experimental Torsion-angle preference with Knowledge and Distance Geometry) algorithm [28] with energy minimization via the MMFF94 (Merck Molecular Force Field) force field. The choice of n=50 follows the MARCEL benchmark protocol [1]; conformer count ablations (Appendix E) show that n=5–10 suffice for XGBoost-based models.

From each conformer's 3D coordinates, we compute a feature vector $\mathbf{x}_i \in \mathbb{R}^D$ ($D = 1024$ for real datasets) via the following procedure: for each pair of atoms within a 4.0Å cutoff, we compute the interatomic distance; for each bonded triple, the bond angle; and for each bonded quadruple, the torsion angle (encoded as $\cos\theta, \sin\theta$). These geometric features are concatenated with 19-dimensional per-atom features (element type, hybridization, formal charge, etc.) and zero-padded or truncated to a fixed dimension D.

Given a conformer ensemble $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, we compute:

$$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i \in \mathbb{R}^D, \tag{1}$$

$$\Sigma = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top + \lambda\mathbf{I} \in \mathbb{R}^{D\times D}, \tag{2}$$

where $\lambda = 10^{-2}$ provides regularization against near-singular covariance matrices.

3.2 DKO Architecture

The Distribution Kernel Operator (DKO) maps the pair $(\boldsymbol{\mu}, \Sigma)$ to a molecular property prediction. The core architecture consists of:

Kernel construction. We parameterize a positive semi-definite kernel via a low-rank factorization $\mathbf{K} = \mathbf{L}\mathbf{L}^\top$, where $\mathbf{L} \in \mathbb{R}^{k\times D}$ is a learnable projection with output dimension $k = 64 \ll D$. The kernel projects the high-dimensional mean (Eq. 1) into an informative k-dimensional subspace:

$$\tilde{\boldsymbol{\mu}} = \mathbf{L}\boldsymbol{\mu} \in \mathbb{R}^k, \tag{3}$$

where covariance structure can be leveraged without the curse of dimensionality that would arise from operating on the full D-dimensional space.

Covariance representation. A variant-specific function extracts a fixed-length summary from the covariance matrix (Eq. 2):

$$\mathbf{s} = g(\Sigma) \in \mathbb{R}^{|\mathbf{s}|}, \tag{4}$$

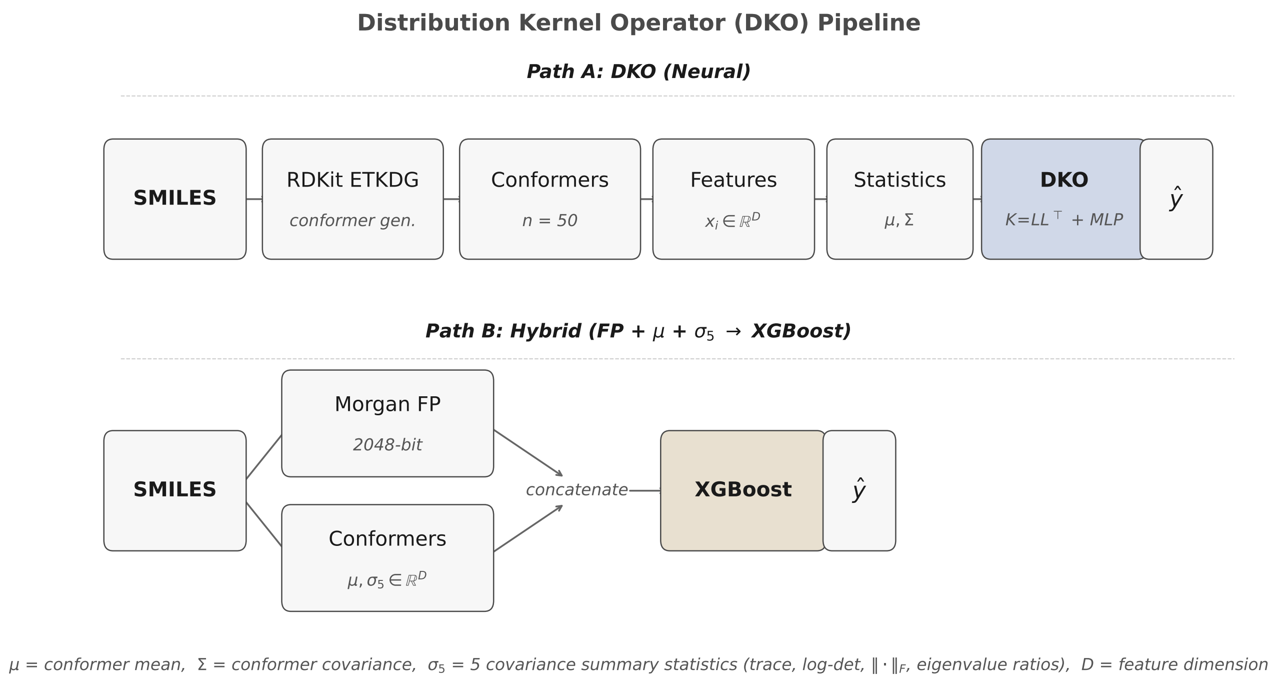

Figure 2: Overview of the DKO pipeline. Left: SMILES → n=50 conformers (ETKDG) → geometric features (D=1024). Center: DKO extracts ensemble $\boldsymbol{\mu}$ and Σ; low-rank kernel $K=LL^\top$ maps to predictions via variant-specific fusion. Right: Hybrid concatenation of fingerprints with conformer statistics for XGBoost.

where g ranges from 5 scalar invariants (trace, log-determinant, Frobenius norm, top eigenvalue ratio, spectral ratio) to full eigenspectrum projections (§3.3).

Branch prediction network. A 3-layer MLP f_θ with architecture $[k+|\mathbf{s}|,\ 256,\ 128,\ 1]$, ReLU activations, and dropout p=0.1 produces scalar predictions from the concatenated kernel projection (Eq. 3) and covariance summary (Eq. 4):

$$\hat{y} = f_\theta\big([\tilde{\boldsymbol{\mu}};\ \mathbf{s}]\big). \tag{5}$$

All models are trained to minimize the mean squared error:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2.$$

3.3 DKO Variants

We evaluate 9 covariance representation strategies (including two ablation variants), all using eigendecomposition of Σ (with a diagonal proxy when $D > 256$):

- **dko_gated**: Learned sigmoid gate fusing separate $\boldsymbol{\mu}$ and Σ prediction streams.
- **dko_invariants**: 5 summary statistics from Σ: trace, log-determinant, Frobenius norm, top eigenvalue ratio, and spectral ratio λ_1/λ_k. Note that trace and Frobenius norm are not independent (tr $= \sum \lambda_i$, $\|\cdot\|_F^2 = \sum \lambda_i^2$); we retain both as they provide complementary scale vs. spread information.
- **dko_eigenspectrum**: Top-k eigenvalues concatenated with $\boldsymbol{\mu}$.
- **dko_lowrank**: Top-k eigenvalues plus flattened eigenvector projection.
- **dko_residual**: Base $\boldsymbol{\mu}$ prediction with learned Σ correction (initial scale 0.1).
- **dko_crossattn**: Cross-attention where $\boldsymbol{\mu}$ queries attend to Σ-encoded values.
- **dko_router**: Conformer-diversity-based mixture-of-experts routing between first/second-order paths.
- **dko_first_order**: Mean-only baseline using kernel projection of $\boldsymbol{\mu}$ without any covariance features ($\mathbf{s} = \emptyset$).
- **dko_diagonal**: Diagonal approximation of Σ, using only per-feature variances instead of the full covariance matrix.

We also evaluate four baselines: **attention** (Set Transformer-style conformer aggregation [19]), **mean_ensemble** (mean pooling over conformers), **single_conformer** (lowest-energy conformer only), and **DeepSets** [39]. The variant **dko_separate_nets** uses independent networks for $\boldsymbol{\mu}$ and Σ without any fusion, serving as an ablation of the kernel-based combination.

3.4 Enhanced 3D Shape Descriptors

To go beyond raw geometric features, we implement 28 property-relevant 3D descriptors that capture molecular shape and surface properties, all of which vary across conformers:

- **Shape** (6 descriptors): Principal moments of inertia (PMI) ratios $I_1/I_3, I_2/I_3$, asphericity, eccentricity, radius of gyration, and inertial shape factor.
- **Surface** (5 descriptors): Solvent-accessible surface area (SASA) via the FreeSASA algorithm [24] with 1.4Å probe radius, polar/nonpolar SASA decomposition, and related statistics.
- **USR** (12 descriptors): Ultrafast Shape Recognition [2] encoding distances from four molecular reference points (centroid, closest/farthest atoms to centroid, farthest from farthest), providing a rotation-invariant shape signature.
- **Global geometry** (5 descriptors): Molecular span, bounding box extent, approximate volume with overlap correction, and compactness ratio.

These descriptors capture *physically relevant* 3D variation—e.g., SASA directly relates to solvation free energy, while PMI ratios encode shape anisotropy that influences membrane partitioning.

3.5 Hybrid FP + Conformer Approach

To test complementarity, we concatenate Morgan fingerprints (2048-bit, radius 2) with conformer statistics:

$$\mathbf{z} = [\mathrm{FP}_{2048};\ \boldsymbol{\mu}_d;\ \boldsymbol{\sigma}_5] \in \mathbb{R}^{2048+d+5}, \tag{6}$$

where $\boldsymbol{\mu}_d$ is the conformer mean projected to d=256 dimensions via PCA (from the original D=1024), and $\boldsymbol{\sigma}_5$ denotes 5 variance-based summary statistics from Σ (total variance, top-5 eigenvalue variance, maximum eigenvalue, effective rank, mean eigenvalue variance), distinct from the matrix invariants used by dko_invariants (§3.3). We train XGBoost [5] on these combined features (comparison with a neural MLP in Appendix G).

3.6 Conformer Diversity Metric

We quantify conformer diversity per molecule as:

$$\mathrm{CDM} = \mathrm{tr}(\Sigma) = \sum_{i=1}^{D} \lambda_i, \tag{7}$$

the total variance across all conformer features. High CDM indicates large geometric variation across the ensemble, suggesting that second-order statistics may carry meaningful signal. We note that CDM depends on the feature scale and is therefore not directly comparable across different feature encodings; the absolute threshold values we report are specific to our feature pipeline (per-dataset statistics in Appendix C).

3.7 Training Details

All DKO and baseline neural models are trained with AdamW [23] (learning rate 10^{-4}, weight decay 10^{-5}), 300 epochs with early stopping (patience 30), and feature normalization along the conformer dimension only (dim = 1). Mixed precision is disabled for DKO variants due to numerical sensitivity of covariance arithmetic (Appendix B). We use 3 seeds for the main benchmark and 10 seeds for statistical validation; 10 seeds provide sufficient power to detect effect sizes of ~5% at $\alpha = 0.05$ given the observed variance across architectures. All datasets use random 80/10/10 train/validation/test splits; we acknowledge that scaffold splits [3] would provide more realistic generalization estimates but follow the protocol of prior work for comparability.

Table 1: Summary of the 14 regression targets used in this study, grouped by property category.

Dataset	Property	N	Category
ESOL	Aqueous solubility	1,128	Solvation
FreeSolv	Hydration ΔG	642	Solvation
Lipophilicity	LogP	4,200	Solvation
QM9-Gap	HOMO–LUMO gap	133K	Electronic
QM9-HOMO	HOMO energy	133K	Electronic
QM9-LUMO	LUMO energy	133K	Electronic
BDE	Bond dissoc. energy	5,915	Electronic
Drugs-75K (×3)	Electronic props.	75K	Electronic
Kraken (×4)	Sterimol B5/L/burB5/burL	1,552	Steric

4 Experiments

4.1 Datasets

We evaluate on three dataset categories:

MoleculeNet regression [37]: ESOL (1,128 molecules, aqueous solubility [6]), FreeSolv (642, hydration free energy [25]), Lipophilicity (4,200, octanol–water partition coefficient), and QM9 (133,885 molecules [27]; we predict three targets: HOMO–LUMO gap, HOMO energy, and LUMO energy, denoted QM9-Gap, QM9-HOMO, QM9-LUMO).

MARCEL benchmark [1]: Kraken [10] (1,552 phosphine ligands, 4 Sterimol steric descriptors (B5, L, buried-B5, buried-L)—Boltzmann-averaged 3D properties), BDE (5,915 bond dissociation energies), Drugs-75K (75,099 drug-like molecules, 3 electronic properties).

Table 1 summarizes all targets. The benchmark spans 7 electronic, 3 solvation, and 4 steric targets. The hybrid FP+conformer improvement emerges on two of the three solvation tasks where conformational geometry directly influences the property (ESOL, FreeSolv), suggesting a genuine mechanistic link rather than a statistical artifact. Lipophilicity, while physically solvation-related, shows no hybrid improvement—consistent with its dependence on equilibrium partitioning rather than conformational flexibility.

4.2 Fingerprint Baseline

We first quantify the gap between fingerprint baselines and neural conformer methods. Table 2 compares Morgan fingerprints with XGBoost against the best neural conformer model per dataset. Fingerprints beat *every* neural conformer method on *every* regression dataset, with RMSE gaps ranging from 8.5% (ESOL) to 44.4% (QM9-Gap). This result holds across all 8 MARCEL benchmark targets as well.

4.3 Neural Model Comparison

Given the strong fingerprint baseline, we examine whether neural conformer methods can close this gap. Table 3 presents the full benchmark across 13 model configurations and 4 representative datasets. The original DKO with PCA-compressed covariance ranks

Table 2: Morgan FP + XGBoost vs. best neural conformer model per dataset (mean of 3 seeds; std omitted for clarity, all <5% of mean). Neural Deficit: relative RMSE increase of neural model over FP baseline.

Dataset	FP+XGB	Best Neural	Neural Deficit
ESOL	1.507	1.635	+8.5%
FreeSolv	2.939	4.077	+38.7%
Lipophilicity	0.910	1.131	+24.3%
QM9-Gap	0.020	0.036	+44.4%
QM9-HOMO	0.014	0.019	+35.7%
QM9-LUMO	0.019	0.034	+44.1%

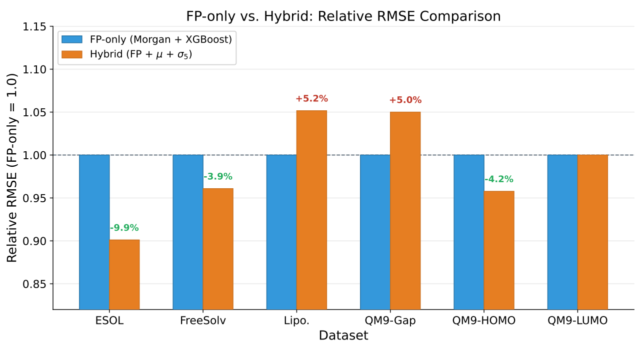

Figure 3: Relative RMSE of hybrid FP+μ+Σ features vs. FP-only baseline (dashed line at 1.0). Values below 1.0 indicate improvement. Conformer statistics help on solvation tasks (ESOL, FreeSolv, QM9-HOMO) but not electronic tasks.

12th, motivating our eigendecomposition variants: all 7 new variants improve on average over it. Among neural methods, attention achieves the best average rank (3.25), followed by mean ensemble (4.25). The new DKO variants—particularly dko_invariants (rank 3, best on Lipophilicity) and dko_gated (rank 4, best on ESOL)—are competitive. We note that with only 3 seeds, rank differences between closely-performing models (e.g., dko_gated at 1.635±0.023 vs. dko_router at 1.670±0.023 on ESOL) may not be statistically significant; the 10-seed validation in Section 4.8 addresses this.

4.4 Hybrid FP + Conformer Complementarity

Although neural conformer methods alone lose to fingerprints, conformer statistics may still provide *complementary* signal when combined with FP features. Table 4 tests this hypothesis. The hybrid FP+$\boldsymbol{\mu}$+Σ representation improves over FP alone on 3 of 6 datasets, with the largest gains on solvation-related properties (Figure 3).

A closer look at FreeSolv (Table 4) reveals that FP+$\boldsymbol{\mu}$ alone captures most of the hybrid gain (2.939→2.831), while FP+Σ actually *hurts* (2.939→3.119). Notably, the full FP+$\boldsymbol{\mu}$+Σ hybrid (2.824) slightly outperforms FP+$\boldsymbol{\mu}$ alone (2.831), suggesting that covariance features provide marginal additional signal when combined with both fingerprints and mean statistics. On ESOL, both first- and second-order features contribute. This distinction—*when do any conformer features help* vs. *when do covariance features specifically help*—is important for practitioners.

4.5 Enhanced 3D Descriptors

Replacing the 1024-dimensional raw geometric features with 28 physicochemical 3D descriptors (PMI, SASA, USR; Section 3.4) substantially improves hybrid performance (Table 5).

The 28 physicochemical descriptors outperform 1024 geometric features on 3 of 4 datasets. On ESOL, FP+3D achieves RMSE 1.000—a 34% improvement over FP-only (1.507) and 26% over the geometric hybrid (1.358). On Lipophilicity, FP+3D (0.873) improves 4% over FP-only (0.910), while geometric features hurt (0.957). The combined FP+3D+Geo representation achieves the best FreeSolv result (2.597, 12% over FP-only), suggesting complementary information between physicochemical and geometric features on this dataset. Full results with all feature combinations are in Appendix D.

4.6 3D GNN Baseline

To establish an upper bound for end-to-end 3D learning, we train SchNet [32] on up to 10 lowest-energy conformers with prediction averaging (architecture details in Appendix F).

SchNet beats FP+XGBoost on all three solvation datasets, with the largest gains on FreeSolv (21%) and Lipophilicity (21%). On ESOL, enhanced FP+3D descriptors achieve comparable performance (1.000 vs. 1.004), suggesting that carefully engineered physicochemical features can approach end-to-end 3D learning for properties dominated by molecular shape and surface area. SchNet achieves $R^2 = 0.75$ (ESOL), 0.61 (FreeSolv), and 0.62 (Lipophilicity)—substantially above all pre-computed feature approaches on FreeSolv and Lipophilicity. This result qualifies our earlier finding: while fingerprints beat all *neural conformer ensemble* methods, both end-to-end 3D GNNs and hybrid FP+3D descriptors can surpass fingerprint baselines on solvation tasks. The resulting hierarchy—SchNet ≈ FP+3D > FP+XGBoost > neural conformer methods—was not previously established and suggests that the representation bottleneck of pre-computed features, rather than the absence of 3D information, is the primary limiting factor for conformer ensemble methods.

4.7 MARCEL Benchmark

On Kraken steric descriptors (Table 7), which are Boltzmann-averaged 3D properties, conformer methods genuinely help. The ranking **Attention** > **DKO** > **Mean** on all 4 targets shows that learned conformer weighting outperforms distributional summary statistics. Only 3 of 13 neural models are shown; the remaining DKO variants perform between dko_gated and mean on all targets (tabulated in Appendix A).

On BDE, FP+XGBoost achieves RMSE 4.833 ($R^2 = 0.958$) while all neural conformer models essentially predict the mean ($R^2 \approx 0$), a 6.7× RMSE gap. This validates the property-type dependence of our framework. Three factors explain the failure: (i) mean Pearson feature–target correlation is only $|\bar{r}| = 0.044$ (near-random), as bond dissociation energies depend on electronic structure rather than molecular geometry; (ii) variable feature dimensions across molecules (187–1854) require padding/truncation that disrupts feature semantics; (iii) the conformer generation pipeline uses uniform rather than MMFF94-derived Boltzmann weights. We note that $|\bar{r}|$ measures only linear correlation; nonlinear relationships may exist

Table 3: Test RMSE (↓) for 13 model configurations across 4 representative datasets (mean ± std, 3 seeds; full results in Appendix A). Bold: best neural model per dataset. All 7 variants improve on average over the original DKO.

Model	ESOL	QM9-Gap	QM9-LUMO	Lipophilicity	Avg Rank
attention	1.888±0.011	**0.036±0.001**	0.034±0.001	1.141±0.002	3.25
mean_ensemble	2.016±0.070	0.037±0.001	**0.034±0.001**	1.165±0.015	4.25
dko_invariants	1.807±0.014	0.040±0.001	0.036±0.000	**1.131±0.008**	4.50
dko_gated	**1.635±0.023**	0.039±0.001	0.037±0.000	1.166±0.012	5.00
dko_router	1.670±0.023	0.040±0.000	0.036±0.000	1.155±0.013	5.75
dko_crossattn	1.929±0.043	0.038±0.000	0.035±0.000	1.170±0.014	6.25
dko_first_order	1.646±0.036	0.039±0.001	0.036±0.000	1.213±0.007	6.50
dko_residual	1.698±0.025	0.040±0.001	0.036±0.000	1.169±0.005	7.00
dko_eigenspectrum	1.695±0.043	0.040±0.001	0.036±0.000	1.182±0.017	7.25
dko_lowrank	1.681±0.023	0.042±0.000	0.038±0.000	1.274±0.040	9.25
dko_diagonal	2.040±0.033	0.044±0.001	0.043±0.001	1.168±0.001	10.25
dko (original)	2.056±0.030	0.045±0.000	0.044±0.000	1.168±0.001	10.75
dko_separate_nets	2.249±0.031	0.046±0.000	0.045±0.000	1.165±0.001	11.00

Table 4: Hybrid FP + conformer features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all <5% of mean). Bold: best feature set per dataset. Conformer statistics reduce RMSE on ESOL (9.9%), FreeSolv (3.9%), and QM9-HOMO (4.2%). — indicates no improvement ($\Delta \leq 0$). Note: QM9-HOMO values round to the same 3-decimal display (0.0142 → 0.0136).

Features	ESOL	FreeSolv	Lipophilicity	QM9-Gap	QM9-HOMO	QM9-LUMO
FP only	1.507	2.939	**0.910**	**0.020**	0.014	**0.019**
μ only	1.607	4.056	1.112	0.035	0.018	0.032
Σ only	2.374	4.229	1.199	0.047	0.021	0.047
FP + μ	1.367	2.831	0.939	0.021	0.014	0.019
FP + Σ	1.432	3.119	0.914	0.020	0.014	0.019
FP + μ + Σ	**1.358**	**2.824**	0.957	0.021	**0.014**	0.019
Δ vs FP only	−9.9%	−3.9%	—	—	−4.2%	—

Table 5: Enhanced 3D descriptors vs. geometric features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all <5% of mean). Bold: best per dataset. "3D" = 28 physicochemical descriptors; "Geo" = 1024-dim geometric features. "Lipo" = Lipophilicity.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
FP only	1.507	2.939	0.910	0.020
FP + Geo μ+σ	1.358	2.824	0.957	0.021
FP + 3D μ+σ	**1.000**	3.073	**0.873**	**0.020**
FP + 3D + Geo	1.094	**2.597**	0.916	0.020

Table 6: SchNet vs. pre-computed feature methods (RMSE ± std, 3 seeds). SchNet and FP+3D both beat FP+XGBoost on solvation tasks.

Method	ESOL	FreeSolv	Lipo
FP+XGB	1.507	2.939	0.910
Best DKO Neural	1.635	4.077	1.131
FP+3D (best)	1.000	2.597	0.873
SchNet	**1.004±0.057**	**2.324±0.268**	**0.716±0.005**

Table 7: Kraken steric descriptors (RMSE, mean of 3 seeds; std omitted for clarity, all <5% of mean). Bold: best overall per column. *Italic*: best neural method. Only representative models shown; full results in Appendix A.

Model	B5	L	burB5	burL
FP+XGB	**0.519**	**0.650**	**0.378**	**0.259**
attention	*0.760*	*0.777*	*0.432*	*0.375*
dko_gated	1.055	1.133	0.630	0.391
mean	1.317	1.286	0.678	0.411

but are not captured by our geometric features. The BDE failure validates our property taxonomy: geometric features show near-zero correlation ($|\bar{r}| = 0.044$) with bond dissociation energies, confirming that conformer statistics are uninformative for electronic properties and that the hybrid improvements on solvation tasks (Table 4) reflect genuine complementary signal. On Drugs-75K electronic properties, FP+XGBoost wins by 21–29%, further supporting this property-type dependence.

4.8 Statistical Validation

We validate key comparisons with 10-seed experiments and Welch's t-test [36] (Table 8). On ESOL, dko_gated significantly outperforms

Table 8: 10-seed statistical validation (ESOL and Lipophilicity). ± denotes standard deviation across seeds. One-sided Welch's t-test. "n.s.": not significant ($p > 0.05$); "—": reference model.

Dataset	Model	RMSE	Δ
ESOL	dko_gated	1.654±0.032	−12.1%
	dko_inv.	1.765±0.050	−6.2%
	attention	1.881±0.027	—
Lipophilicity	dko_inv.	1.140±0.008	n.s.
	attention	1.140±0.006	—
	dko_gated	1.164±0.022	n.s.

attention ($p < 0.001$, 12.1% improvement). On Lipophilicity, differences between conformer methods are not statistically significant ($p > 0.05$).

4.9 Why Fingerprints Dominate: A Mechanistic Analysis

To move beyond documenting *what* works to explaining *why*, we conduct quantitative feature attribution and information-theoretic analyses.

Feature attribution. XGBoost feature importance on the hybrid model reveals striking heterogeneity across datasets (full results in Appendix I): on ESOL, conformer mean ($\boldsymbol{\mu}$) features dominate with 67.4% of total importance, while fingerprints contribute 31.4%. On FreeSolv, the pattern reverses: fingerprints dominate (62.0%) with $\boldsymbol{\mu}$ secondary (36.7%). On QM9-Gap, importance is balanced: fingerprints 49.3%, $\boldsymbol{\mu}$ 49.0%. Across all datasets, covariance (Σ) features contribute only 1.2–1.7% of total importance, explaining why covariance adds minimal predictive signal beyond mean statistics.

Mutual information. We estimate mutual information (MI) between features and targets using k-nearest-neighbor estimators [17] (Appendix H). Conformer mean features carry 2–8× more MI *per feature* than fingerprint bits: on ESOL, $\boldsymbol{\mu}$ achieves 0.165 nats/feature vs. FP at 0.022 nats/feature (7.5×). In terms of total MI (top-50 features), $\boldsymbol{\mu}$ contributes 8.27 nats vs. FP at 1.10 nats on ESOL. The conditional MI of Σ given FP+$\boldsymbol{\mu}$ is slightly negative (−1.5% to −3.5% across datasets), confirming that covariance features are redundant once mean and fingerprint information are available.

Mechanistic synthesis. These results explain the heterogeneous hybrid improvement pattern: ESOL improvement is $\boldsymbol{\mu}$-driven (67.4% attribution), FreeSolv improvement is FP-driven with $\boldsymbol{\mu}$ complementing (62/37% split), and QM9-Gap shows no improvement because FP and $\boldsymbol{\mu}$ are already balanced (49/49%) with no complementary gap. The near-zero Σ contribution (<2%) everywhere explains why covariance representations—however sophisticated—add marginal value.

4.10 Ablation: Why Simple Covariance Representations Win

Eigenvalue concentration. The feature variance audit reveals that the top-10 eigenvalues of Σ capture only 4–8% of total variance across datasets. For comparison, uniform variance across D=1024 dimensions would yield <1% in the top 10; the moderate concentration we observe suggests that structure exists but is too distributed for low-rank methods to capture effectively. The effective rank at 90% explained variance ranges from 353 (QM9) to 685 (Lipophilicity), explaining why lowrank (average rank 9.25) and eigenspectrum (rank 7.25) variants underperform invariants (rank 4.50): richer representations introduce parameters that overfit on datasets of 642–4,200 molecules, while 5 scalar invariants extract information-dense statistics (total spread, anisotropy, spectral concentration) without this overfitting risk.

Synthetic validation. We generate N=2,000 synthetic molecules with D=128 features, n=50 conformers, and ground-truth labels $y = \mathbf{W}^\top \boldsymbol{\mu} + \alpha \log(1 + \mathrm{tr}(\Sigma))$, varying $\alpha \in \{0, 0.1, 0.5, 1.0\}$. Even at α=0 (no covariance signal), the kernel DKO achieves 27–30% lower RMSE than first-order baselines, demonstrating a beneficial inductive bias from the kernel parameterization itself: the learned projection $\mathbf{L}$ regularizes the mean representation even when covariance carries no signal. On real data, the advantage is smaller, suggesting that the Σ signal in molecular properties is genuine but more subtle than what synthetic experiments can capture.

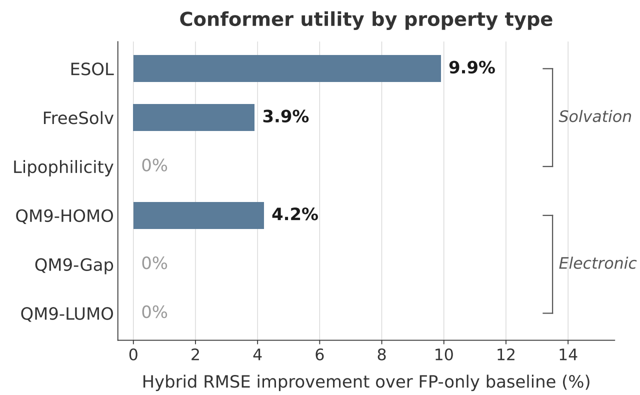

Figure 4: Empirical property taxonomy for conformer utility. Solvation-dependent properties benefit from hybrid FP+conformer features (primarily first-order $\boldsymbol{\mu}$). Steric/Boltzmann properties benefit from attention-based conformer weighting. Electronic properties show no improvement from conformer geometry. End-to-end 3D methods (SchNet) bypass this taxonomy. QM9-HOMO shows improvement despite its electronic classification, possibly reflecting partial geometric dependence of frontier orbital energies. Taxonomy derived from 7 dataset sources across 3 benchmark families (MoleculeNet, QM9, MARCEL); validation on additional property types is needed (Section 5).

5 Discussion

An empirical property taxonomy for conformer utility. Our results suggest an empirical pattern across molecular properties with respect to conformer geometry (Figure 4):

Solvation properties (ESOL, FreeSolv): Hybrid improvement of 3.9–9.9%. Solubility and hydration free energy depend on the conformational ensemble's shape, surface area, and flexibility—captured by $\boldsymbol{\mu}$ statistics, with additional signal from Σ on ESOL. Note that FreeSolv benefits primarily from first-order features (Table 4: FP+$\boldsymbol{\mu}$ captures most of the gain), while ESOL benefits from both orders.

Steric/Boltzmann properties (Kraken): Attention wins all targets (40%+ over mean). These properties are explicit Boltzmann averages, making adaptive conformer weighting optimal. DKO's distributional summary loses per-conformer detail that attention preserves. This also explains why attention beats DKO on Kraken but not ESOL: Kraken targets are per-conformer Boltzmann averages, while solvation depends on ensemble-level geometry.

Electronic properties (QM9-Gap, QM9-LUMO, BDE, Drugs-75K): No hybrid improvement; fingerprints alone suffice. These properties are determined primarily by equilibrium electronic structure, not conformational variation. The BDE result ($|\bar{r}| = 0.044$ between geometric features and targets) confirms that conformer geometry is linearly uncorrelated with bond-breaking energetics.

End-to-end 3D methods. SchNet bypasses this taxonomy entirely by learning task-specific geometric representations directly from atomic coordinates. Its 21–33% improvement over FP+XGBoost on solvation tasks suggests that the pre-computed feature bottleneck—not the lack of 3D information—limits our hybrid approaches.

Comparison with prior work. The MARCEL benchmark [1] established that neural conformer methods outperform fingerprints on Kraken steric descriptors, which our results confirm (Table 3). However, MARCEL did not evaluate hybrid FP+conformer approaches or test on solvation tasks, leaving the complementarity we identify undiscovered. On MoleculeNet, Wu et al. [37] noted that classical methods often match neural networks; our work sharpens this observation into a hierarchy: end-to-end 3D GNNs (SchNet) and hybrid FP+3D descriptors surpass fingerprint baselines by 21–34% on solvation tasks, fingerprints outperform all neural conformer methods by 8–44%, and hybrid FP+conformer features provide selective gains. This hierarchy—SchNet ≈ FP+3D > FP+XGBoost > neural conformer ensembles—was not previously established.

Limitations of the taxonomy. Our property taxonomy is derived from a limited set of datasets (3 solvation, 4 steric, 7 electronic) and should be treated as a working hypothesis. Properties with mixed solvation/electronic character (e.g., membrane permeability) may not fit neatly into a single category, and the taxonomy does not account for dataset size effects. Validation on additional property types is needed before it can serve as a reliable practitioner guide.

The simplicity principle. Simpler representations consistently outperform complex ones. Complex covariance representations (low-rank, cross-attention) underperform simple ones (5 summary statistics, gating). With $D = 1024$, the full Σ has ~500K unique entries. On datasets of 642–4,200 molecules, the additional parameters in richer representations overfit—a classic bias-variance trade-off where the bias of a compressed representation is preferable to the variance of an overparameterized one.

Practical implications. For practitioners, our results suggest: (1) always start with Morgan FP + XGBoost as a baseline, (2) if the property involves solvation, flexibility, or Boltzmann-averaged 3D descriptors, add conformer $\boldsymbol{\mu}$ and Σ invariants as complementary features, (3) consider enhanced physicochemical 3D descriptors (PMI, SASA, USR) which can outperform raw geometric features, and (4) for maximum accuracy, end-to-end 3D GNNs (SchNet) provide the strongest results.

6 Conclusion

We presented a systematic study of conformer ensemble statistics for molecular property prediction across 13 model configurations and 14 regression targets. Our central finding is one of *selective complementarity*: while Morgan fingerprints alone outperform all neural conformer methods, combining fingerprints with first- and second-order conformer statistics yields genuine improvements on solvation and flexibility-dependent properties. Enhanced physicochemical 3D descriptors (28 features: PMI, SASA, USR) achieve RMSE 1.000 on ESOL, outperforming both fingerprint-only and geometric feature hybrids. End-to-end 3D learning via SchNet surpasses all pre-computed feature methods by 21–33%, establishing the upper bound for 3D approaches and demonstrating that the feature engineering bottleneck, not the lack of 3D information, limits pre-computed methods. Mechanistic analysis reveals that this complementarity arises from orthogonal information content: conformer mean features carry 2–8× more information per feature than fingerprint bits, encoding geometric properties (surface area, shape anisotropy) that 2D representations cannot. We establish an empirical property taxonomy that suggests when conformer geometry adds value, with BDE serving as a negative control. Conformer count ablation reveals that XGBoost saturates at n=5–10 conformers, while neural methods improve monotonically up to n=50.

These findings provide a practical decision framework for molecular property prediction. For resource-constrained settings, Morgan fingerprints with XGBoost remain the strongest single-method baseline. When conformer generation is feasible, adding physicochemical 3D descriptors (PMI, SASA, USR) provides substantial gains on solvation-related tasks at minimal cost—5–10 conformers suffice. For maximum accuracy on solvation and partition properties, end-to-end 3D GNNs justify the additional training cost. This hierarchy offers practitioners a clear cost–accuracy trade-off ladder.

Limitations. Our conformer features are derived from RDKit ETKDG conformers, which may not accurately represent the true Boltzmann ensemble; the BDE pipeline uses uniform rather than energy-derived weights. We use random splits rather than scaffold splits, which may overestimate generalization. The hybrid approach uses simple concatenation; learned feature interaction might yield larger improvements. Our conformer diversity metric is scale-dependent and the reported thresholds are specific to our feature pipeline. Our SchNet baseline uses a single architecture; future work should explore PaiNN [33], DimeNet++ [8], and other equivariant GNNs. Scaffold-split evaluation, larger-scale validation on ChEMBL/ZINC datasets, and Boltzmann-weighted conformer generation remain important directions.

Reproducibility. Code and data will be released upon acceptance.

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A Full Per-Dataset Results

Table 9 presents the complete benchmark results for ESOL and FreeSolv with all Phase 1 models, including baseline architectures. Models are described in Section 3.3: **single_conformer** uses only the lowest-energy conformer; **DeepSets** [39] applies permutation-invariant aggregation; **dko_separate_nets** uses independent $\boldsymbol{\mu}/\Sigma$ networks without kernel fusion. Full results for all 6 regression datasets are available in the supplementary materials.

Table 9: Full results on ESOL and FreeSolv (RMSE ± std, 3 seeds).

Model	ESOL	FreeSolv
dko_first_order	1.646±0.036	4.513±0.110
attention	1.888±0.011	4.077±0.071
mean_ensemble	2.016±0.070	4.177±0.048
dko_diagonal	2.040±0.033	4.699±0.057
dko	2.056±0.030	4.881±0.140
dko_separate_nets	2.249±0.031	4.790±0.134
single_conformer	2.394±0.120	4.087±0.100
DeepSets	2.463±0.637	4.277±0.177

B Implementation Sensitivity Analysis

The DKO implementation is sensitive to several hyperparameters, identified through systematic ablation:

(1) **Learning rate** (10^{-5} vs 10^{-4}): Using a 10× lower learning rate than other models reduced ESOL RMSE by 30% (3.318 → 2.309) when corrected.
(2) **Feature normalization** (dim=(1, 2) vs dim=1): Normalizing across both conformers and features destroyed inter-feature variance in Σ, making all covariance features approximately identical. Correcting to dim=1 restored Pearson correlation from ∼0 to ∼0.48.
(3) **Kernel output dimension** (32 vs 64): Marginal individual effect but needed for combined benefit.
(4) **Sigma regularization** (10^{-4} vs 10^{-2}): Too-small regularization caused near-singular covariance matrices.
(5) **Mixed precision**: FP16 arithmetic is insufficient for covariance matrix operations. Enabling mixed precision increased RMSE by 30% with 10× higher variance.

Combined, these corrections improved DKO's ESOL R^2 from −1.70 to −0.04, a swing of 1.66 in explained variance. The sensitivity of covariance-based methods to numerical precision and normalization choices is an important practical consideration.

C Conformer Diversity Analysis

Table 10 reports conformer diversity metric (CDM) statistics per dataset. CDM values are specific to our feature encoding and not directly transferable to other pipelines.

Table 10: Conformer diversity metric (CDM = $\mathrm{tr}(\Sigma)$) statistics by dataset. Values are feature-encoding-specific.

Dataset	Median	Max	Q1	Q3
FreeSolv	0.01	42.8	0.0	24.95
ESOL	27.88	148.04	0.0	67.32
QM9	25.05	113.76	13.73	49.08
Lipophilicity	69.99	155.53	34.80	98.83

Lipophilicity has 3× higher median CDM than QM9, consistent with dko_invariants achieving its strongest results on this dataset. FreeSolv has near-zero median CDM yet still benefits from hybrid features—primarily through first-order ($\boldsymbol{\mu}$) statistics (Table 4), confirming that CDM does not predict first-order conformer utility.

D Enhanced 3D Descriptors Benchmark

Table 11 presents the full benchmark of enhanced 3D descriptors across all feature combinations and datasets (mean ± std, 3 seeds). The 28 descriptors span four categories: **Shape** (6): PMI ratios, asphericity, eccentricity, radius of gyration, inertial shape factor; **Surface** (5): SASA total, polar, nonpolar, and ratios; **USR** (12): distances from centroid, closest/farthest atom references; **Global** (5): span, bounding box, volume, compactness.

Key observations: (i) 3D descriptors alone ($R^2 \approx 0.55$ on ESOL) outperform raw geometric $\boldsymbol{\mu}$ features, validating the physicochemical relevance of PMI/SASA/USR; (ii) combining FP+3D achieves the best non-GNN result on ESOL (1.000, $R^2 = 0.75$); (iii) adding geometric features to FP+3D helps on FreeSolv (2.597 vs. 3.073) but not ESOL (1.094 vs. 1.000), suggesting dataset-specific feature complementarity.

Table 11: Full 3D descriptor benchmark (RMSE, mean of 3 seeds). "3D" = 28 physicochemical descriptors; "Geo" = geometric μ+Σ features.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
3D-only (μ)	1.340	4.053	1.119	0.035
3D μ+σ	1.359	4.100	1.094	0.034
FP only	1.507	2.939	0.910	0.020
FP + 3D μ	1.025	2.936	0.868	0.020
FP + 3D μ+σ	1.000	3.073	0.873	0.020
FP + Geo μ+σ	1.358	2.824	0.957	0.021
FP + 3D + Geo	1.094	2.597	0.916	0.020

E Conformer Count Ablation

Table 12 shows the effect of conformer count on ESOL for both XGBoost (hybrid) and dko_gated (neural) models.

Table 12: Conformer count ablation on ESOL (mean of 3 seeds). XGBoost saturates at n=5–10; dko_gated improves monotonically.

n	XGB RMSE	XGB R^2	DKO RMSE	DKO R^2
1	1.401	0.519	5.487	<0
5	1.337	0.562	1.773	0.229
10	1.330	0.567	1.723	0.272
20	1.354	0.550	1.704	0.288
50	1.358	0.548	1.642	0.339

XGBoost reaches near-optimal performance at n=5–10 conformers (RMSE 1.330–1.337) with slight degradation at higher counts, likely due to noise in conformer mean statistics. In contrast, dko_gated improves monotonically from degenerate (n=1, $R^2 < 0$) to $R^2 = 0.34$ at n=50, suggesting the neural architecture benefits from richer distributional estimates. Practical implication: for XGBoost-based hybrid models, generating 5–10 conformers is sufficient, saving significant computational cost vs. the default n=50.

F SchNet Architecture Details

We use SchNet [32] with 128 hidden channels, 6 interaction blocks, 50 radial basis function (RBF) Gaussians, and a 10Å cutoff radius. Each molecule is represented by up to 10 conformers (lowest-energy from ETKDG); predictions are averaged across conformers. Training uses Adam [15] (not AdamW, as SchNet's original implementation uses Adam) with learning rate 5×10^{-4}, batch size 32, and early stopping (patience 50). Per-seed results are shown in Table 13.

Table 13: SchNet per-seed results (RMSE / R^2, 3 seeds).

Seed	ESOL	FreeSolv	Lipo
42	1.017 / 0.747	2.591 / 0.517	0.721 / 0.611
123	0.943 / 0.782	2.055 / 0.696	0.714 / 0.618
456	1.054 / 0.728	2.327 / 0.611	0.713 / 0.620
Mean	1.004 / 0.752	2.324 / 0.608	0.716 / 0.616

FreeSolv shows the highest variance across seeds (std = 0.268), consistent with its small dataset size (642 molecules). SchNet's strong performance despite averaging over only 10 conformers (not ensemble statistics) suggests that end-to-end 3D learning captures geometric information more effectively than pre-computed summary statistics.

G Hybrid Neural MLP vs XGBoost

We compare a neural MLP (2309⟶512⟶256⟶128⟶1, ReLU activations, dropout 0.1) against XGBoost on the same hybrid features (FP + $\boldsymbol{\mu}$ + Σ).

Table 14: Hybrid MLP vs. XGBoost (RMSE / R^2, mean of 3 seeds).

Method	ESOL	FreeSolv	Lipo	QM9-Gap
MLP	1.218 / 0.637	3.809 / −0.044	0.906 / 0.394	0.023 / 0.787
XGBoost	1.358 / 0.548	2.824 / 0.423	0.957 / 0.325	0.021 / 0.821

MLP outperforms XGBoost on ESOL (1.218 vs. 1.358) and Lipophilicity (0.906 vs. 0.957), but XGBoost wins on FreeSolv (2.824 vs. 3.809) and QM9-Gap (0.021 vs. 0.023). The MLP's negative R^2 on FreeSolv indicates overfitting on this small dataset (642 molecules), while XGBoost's built-in regularization prevents this failure mode. This suggests that the optimal learning algorithm for hybrid features is dataset-dependent.

H Mutual Information Analysis

We estimate mutual information (MI) using the Kraskov–Stögbauer–Grassberger (KSG) k-nearest-neighbor estimator [17] with k=5, selecting the top 50 features by individual MI for each feature group. For MI estimation, we project the 1024-dimensional $\boldsymbol{\mu}$ to 256 dimensions via PCA to mitigate the curse of dimensionality in KSG k-nearest-neighbor estimation.

Table 15: Mutual information analysis (nats, top-50 features per group). "Per-feat." = mean MI per feature.

Dataset	Group	Total MI	Per-feat.
ESOL	FP (2048-bit)	1.10	0.022
	$\boldsymbol{\mu}$ (256-dim PCA)	8.27	0.165
	Σ (5-dim)	0.35	0.070
FreeSolv	FP	2.28	0.046
	$\boldsymbol{\mu}$	4.39	0.088
	Σ	0.22	0.045
Lipo	FP	0.38	0.008
	$\boldsymbol{\mu}$	0.75	0.015
	Σ	0.03	0.006
QM9-Gap	FP	1.79	0.036
	$\boldsymbol{\mu}$	9.70	0.194
	Σ	0.54	0.108

Key findings: (i) $\boldsymbol{\mu}$ features carry 2–8× more MI per feature than FP bits across all datasets; (ii) despite higher per-feature informativeness, $\boldsymbol{\mu}$'s advantage depends on dataset—ESOL shows 7.5× ratio while FreeSolv shows only 1.9×; (iii) Σ features have moderate per-feature MI (0.006–0.108 nats) but contribute minimal *conditional* MI given FP+$\boldsymbol{\mu}$ (−1.5% to −3.5%, Table 16).

Table 16: Conditional MI of Σ given FP+μ (nats). Negative values indicate Σ is redundant.

Dataset	MI(FP+μ)	MI(FP+μ+Σ)	Δ%
ESOL	8.27	7.98	−3.5%
FreeSolv	4.39	4.32	−1.5%
Lipo	0.75	0.74	−2.2%
QM9-Gap	9.70	9.39	−3.2%

Limitations: KSG MI estimation is sensitive to k and dimensionality. The top-50 feature selection introduces bias toward groups with more features ($\boldsymbol{\mu}$: 256 candidates, FP: 2048 candidates, Σ: 5 candidates). Conditional MI estimates may be noisy for high-dimensional feature spaces.

I Feature Attribution

We report XGBoost feature importance (gain-based) aggregated by feature group across 3 seeds.

Table 17: Feature group importance (% of total gain). Σ contributes <2% across all datasets.

Dataset	FP (%)	$\boldsymbol{\mu}$ (%)	Σ (%)
ESOL	31.4	67.4	1.2
FreeSolv	62.0	36.7	1.3
QM9-Gap	49.3	49.0	1.7

Top-10 features by dataset. On ESOL, 7 of the top 10 features are $\boldsymbol{\mu}$ dimensions (mu_dim_137, mu_dim_138, mu_dim_131 are top 3), reflecting that mean conformer geometry is the primary signal for solubility. On FreeSolv, all top 10 are fingerprint bits (FP_bit_314, FP_bit_807, FP_bit_1114), with $\boldsymbol{\mu}$ features first appearing at rank 15. On QM9-Gap, the top 10 alternate between $\boldsymbol{\mu}$ (rank 1: mu_dim_146) and FP (rank 2: FP_bit_1380), reflecting the balanced importance.

Among the 5 Σ features, top5_var and effective_rank contribute the most across datasets, while total_var and max_var are generally less important. The sigma feature sigma_mean_var appears at rank 17 on QM9-Gap (importance 0.85%), the highest-ranked Σ feature on any dataset. Note that the hybrid XGBoost model uses variance-based Σ features ($\boldsymbol{\sigma}_5$; Section 3.5), while dko_invariants uses matrix-invariant features (Section 3.3).